ECE250: Signals & Systems Quiz-11

Date: 26th November, 2025 **Total Marks:** 10 **Duration:** 15 Minutes

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Note:

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
2. Institute Plagiarism policy are strictly applicable.

[CO2, CO5] [Q1]: Consider the following system functions for stable LTI systems. For each case 1) plot the ROC, 2) find the impulse response h[n], and 3) determine whether or not the corresponding system is causal.

(a) **[5 Marks]**

$$H(z) = \frac{1 - \frac{4}{3}z^{-1} + \frac{1}{2}z^{-2}}{z^{-1}(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})}$$

(b) **[5 Marks]**

$$H(z) = \frac{z - \frac{1}{2}}{z^2 + \frac{1}{2}z - \frac{3}{16}}$$